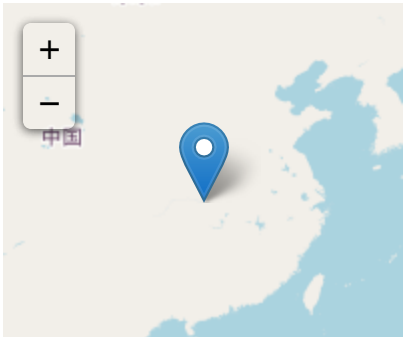
AquaPedia Case Study Database

China: The Three Gorges Dam Hydroelectric Project

Case Description



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Geolocation:	30° 45' 7.94" N, 111° 16' 1.92" E
Climate Descriptors:	Humid mid-latitude (Köppen C-type)
Predominent Land Use Descriptors:	agricultural- cropland and pasture
Important Uses of Water:	Hydropower Generation
Water Features:	Yangtze River;
Riparians:	China;
Water Projects:	Three Gorges Dam

Summary

The Yangtze River is the largest river in China. It is also the third largest river in the world, after the Nile and the Amazon, flowing for 6,300 kilometers (3,915 miles), about twice the length of the US-Mexico border. The Three Gorges Dam (TGD) is an ambitious hydro-industrial complex consisting of a 2,309-meter-long, 185-meter-high dam, a five-tier ship lock, and 32 hydropower turbo-generators that aims to generate a total electric capacity of 22,500 MW to reduce China's dependency on coal-fired generation at the same time it provides a clean source of energy. The project was intended to contribute to flood reduction and flood control in the Yangtze River banks, facilitate navigation upstream and stimulate tourism and trade in the region.

Despite its tremendous hydropower potential, experts assert that the TGD will not be able to operate at full capacity. In the end, the hydropower solution has proven to be not as efficient as the Chinese authorities expected. Operationally it would require 175 meters depth in the dam, but in order to guarantee flood control, the water level of the dam will have to be reduced to 145 meters before each

flood season to provide up to 22.1 billion m3 of flood control capacity.[1] Increased water levels for navigation and sediment formation problems have prevented the TGD and other dams in China (such as the Sanmexia Dam) to operate at full capacity. It is estimated that 140 towns, 326 townships and more than 1,500 villages belonging to 18 different counties were partially or completely flooded after the TGD completion.[2] The inundated area covered about 1,282 cultural heritage places. In 2007, the Chinese government announced that "at least 4 million people in the area were to be relocated to cities in the next 10 to 15 years" independent estimates show this figure could go up to a total of 6 million people to be resettled due to the dam construction or its environmental impacts.[3]

Due to the problems and consequences mentioned above, some specialists point out that the TGD is likely to reduce its estimated proportion of hydropower energy generation by the year 2020 from a previously estimated 40% further down to 26% in order to remedy of the situation at hand. [4]

Natural, Historic, Economic, Regional, and Political Framework

The Three Gorges Dam (TGD) project was long coveted by the Chinese government. The first intention to build the dam was expressed by Sun Yat-sen in 1919. In 1954 severe flooding along the river left 30,000 people dead and one million people homeless. Mao Zedong backed the project to address the constant threat of flooding an ambition he also expressed in one of his most famous poems "Swimming"[5] (1956) revealing his intentions to build the dam. Dams were considered a landmark of development embraced by the highest level government officials and personally endorsed by Premier Li Peng , being himself a hydroelectric engineer.

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A Closed decision-making process

Premier Li Peng advocated for the construction of the dam, influenced by a nationalistic vision of modernization, arguing the project would allow flood control and would provide a readily available alternative energy source for power generation. He was supported by a pro-dam faction in the National People's Congress (NPC) led by Qian Zhengying, the Minister of Water Resources and Electrical Power. Deliberations in Congress were driven by technically illiterate supporters providing unrealistic budgetary figures that seriously underestimated true costs and ignored potential dangers while exaggerating benefits.[6] To illustrate this fact, while deliberating on the construction of the TGD, among the panel of 412 experts involved in the dam approval there was not any sociologist, cultural anthropologist or archaeologist to discuss the assessment of the cultural heritage affected by the reservoir impoundment. [7]

The decision was broadly based on projected benefits: (1) Energy generation.- Hydropower was regarded as the ultimate solution to China's dependency to coal and transition from coal-burning power-plants to hydropower was aimed to reduce greenhouse emissions and satisfy growing demand of energy; (2) Security.- To achieve flood reduction and flood control; (3) Trade.- Increased water depth would benefit upstream navigation to about 80% of China's total navigable cargo.

After more than 70 years of debate the project was approved in 1992. The vote by the 2,600 legislators generated 177 nays and 644 abstentions, an unprecedented demonstration of opposition to a project for which the government had lobbied heavily. Opposition emphasized growing concerns on technical issues and scientific uncertainty on the expected outcomes, as a project of this scope was never put in place before.[3]

A Timeline of Events[3]

1919	First mention of the Three Gorges dam project in Sun Yat-sen's "Plan to Develop Industry."
1931	Massive flooding along the Yangtze River kills 145,000 people.
1932	Nationalist government proposes building a low dam at Three Gorges.
1935	Massive flooding kills 142,000 people.
1940s	The U.S. Bureau of Reclamation helps Chinese engineers identify a site.
1947	Nationalist government terminates all design work.
1949	Communist revolution in China.
1953	Mao Zedong proposes building a dam at Three Gorges to control flooding.
1954	Flooding along the Yangtze leave 30,000 people dead and one million people homeless.
1955	Soviet engineers play a role in project planning and design.
January 1958	Mao appoints Zhou Enlai to begin planning along Yangtze.
May 1959	Yangtze Valley Planning Office (YVPO) identifies Sandouping site for dam.
1966	All work halted by the Cultural Revolution (1965–1975).
1976	Planning recommences.
February 1984	Ministry of Water Resources and Electric Power recommends immediate commencement of construction.
Spring 1985	The National People's Congress delays a decision until 1987 because of economic difficulties.
1986	The Chinese Ministry of Water Resources and Electric Power asked the Canadian government to finance a feasibility study
August 1988	Canadian-World Bank "Three Gorges Water Control Project Feasibility Study" is completed and recommends construction at "an early date."
February 28, 1989	Dai Quing Book Yangtze! Yangtze! released.
February 1992	Politburo Standing Committee agrees to the construction of the project.
April 3, 1992	China's National People's Congress (NPC) formally approved the "Resolution on the Construction of the Yangtze River Three Gorges Project." 177 delegates oppose the project, 644 abstain, 1,767 approve.
April 27, 1992	The Canadian government cancels development assistance for the project.
May 1992	179 members of the Democratic Youth Party reportedly detained in connection with their protests against the Three Gorges project in Kai County, Sichuan (HRW 1995).
January 1993	An armed fight involving over 300 persons occurred in the vicinity of the dam (HRW 1995).
December 14, 1993	The U.S. Bureau of Reclamation terminates agreements for technical services because of economic and environmental impacts.
Early 1994	The full resettlement program begins in earnest.
December 14, 1994	Premier Li Peng formally declared the project under construction.
May 1996	The US Ex-Im Bank's board votes unanimously to withhold support for the project and voices serious reservations about the dam's environmental and social impacts and its economic viability.
August 1997	China awards a contract for 14 power generating units to GEC Alsthom, ABB, and an industrial consortium formed by Germany's Voith and Siemens and General Electric Canada (VGS).

September 1997	The State Development Bank of China signs a loan package with Germany's Kreditanstalt Fur Wiederaufbau, Dresdner Bank, and DG Bank that includes both export credits and a $200 million commercial loan.
November 1997	Yangtze River dammed
May/June 2003	Dam is finished and the first water is impounded. The level of water in the reservoir rises to 135 meters in June.
July 2003	The first electricity is produced.
June 2006	Dam is completed.
October 2006	North side generators reach full capacity. Water level reached 156 meters.
2007	14 700-MW turbine generators in operation.
2008	New estimated completion date.
2009	Project Completion. Reservoir estimated to be raised to 175 meters.
July 2012	Operation started with 32 turbine generators.

Issues and Stakeholders

Unpredictability, limited knowledge and knowledge representation (models), and lack of information (data)

NSPD: Water Quantity, Water Quality, Ecosystems
Stakeholder Types: Federated state/territorial/provincial government, Sovereign state/national/federal government

Environmental Uncertainty

Past experience in China shows that unpredictable climatological events cannot be foreseen, but measures could have been taken to manage catastrophic events in a better way. In 1975 a devastating typhoon (a phenomenon expected once every 100 years) hit Henan province causing the water level of existing dams to exceed the spillway capacity overflowing and destroying its structure. At that time, the dam at the Banqiao Reservoir already exceeded its capacity (originally designed with a capacity of 492 million cubic meters, but had to accommodate more than 697 million cubic meters of floods) and after the typhoon stroke it collapsed. Downstream dikes and flood diversion projects could not resist such a deluge provoking a chain reaction that crumbled 62 dams overnight. This tragedy caused an estimated 85,000 people to die. Two million more were trapped for weeks, and some 11 million were stricken by disease and famine in the aftermath.[8] This experience showed the vulnerability to unforeseen events such as Typhoons and the impact they have on water and dam management in terms of the lack of appropriate management with an early warning system or a widely publicized evacuation plan that contributes to save thousands of lives in the event of unavoidable unpredictability.

Seismic Risk

Seismological analysis of the reservoir reveal the TGD is situated over an active fault line and the weight of the water could trigger an earthquake that could demolish the dam. Also, fluctuating water levels could cause instability on the riverbanks resulting in landslides and tidal waves. Advocates of the TGD claim that it is designed to withstand any seismic shock and that most of the Yangtze shoreline is solid rock. [9]

A 2010 study by seismologists at the China Earthquake Administration confirmed that the TGD has significantly increased seismic activity along the dam's reservoir. Seismic monitors in Hubei Province registered 3,429 earthquakes between June 2003 (when inundation of the reservoir began) and December 2009. Some specialists suggest that the shifting weight of the reservoir may have triggered the May 2008 quake in Sichuan Province, in which least 87,000 people died.

The Flood - Sedimentation - Power Generation Paradox

Historical records show that floods occurred in the Yangtze at about one per decade.[2] In order to play its claimed role in flood control, the water level of the TGD will have to be reduced to 145 meters before each flood season to provide up to 22.1 billion m^3 of flood control capacity. Full capacity electric power generation would require 175 meters depth.[2] The Chinese government was confident that keeping this level will guarantee flood control in the region, but evidence from similar projects has shown the opposite (Sanmenxia Dam).

To remove sediment from the TGD, the reservoir level has to be lowered to 145 meters during the flood season (from May to September), when the water carries most of its sediment load; this would allow to remove the silt through sluice gates and bottom outlets. Then from October to April (dry season), the reservoir level will be raised again to 175 m.[10] But, in the event of heavy rains upstream the dam during the wet season, discharge gates should be closed to prevent floods downstream. But this will cause the build-up of large amount of silt behind the dam which will paradoxically cause the floods it aims to prevent downstream. As heavy rain and typhoons are extremely difficult to predict, there is general agreement among specialists that the TGD will not provide the effective control for the floods envisioned by the Chinese Government.

Increasing the depth of the reservoir for upstream navigation and full capacity electric generation of 22,500 megawatts at 175 m. will contribute to the buildup of sediment, causing the dam to lose flood control capability (as silt is drained from the dam during the rainy season). On the other hand, to keep the water level at 145m will hinder navigation upstream (due to lower water level), and will increase the amount of land that must be flooded.[11]

Water Contamination and Toxic Waste

NSPD: Water Quality, Ecosystems
Stakeholder Types: Federated state/territorial/provincial government, Sovereign state/national/federal government, Local Government, Environmental interest, Community or organized citizens

The filling of the reservoir to 175 meters flooded more than 1,300 factories and mines, including about 4,000 hospitals, around 40,000 graveyards, and about 200 garbage dumping sites, with the subsequent migration of toxics including arsenic, sulfides, cyanides and mercury from these sources to the reservoir water. Not only industrial and domestic pollutants may hazard the aquatic biodiversity; but also the massive use of fertilizers by farmers causes the release of nitrates and phosphates into the dam. These calculations were seriously overlooked ex-ante and had a huge impact ex-post in the contamination of the Yangtze River basin.[12]

Chinese Government - President of China - Premier State Council - Vice President - Communist Party of China - National People's Congress (NPC) - Ministry of Water Resources - Yangtze Valley Planning Office - The Three Gorges Corporation - China Development Bank - Provinces and municipalities local governments.

NGO's and Civil Society - International Rivers - Mountainous Disaster and Environment Institute in Chengdu - Chinese Academy of Sciences (CAS)

- Research Institute for Protection of the Yangtze Water Resources

Resettlement - cultural and archaeological loss - agricultural and livelyhood changes

NSPD: Ecosystems, Governance, Assets, Values and Norms
Stakeholder Types: Federated state/territorial/provincial government, Sovereign state/national/federal government, Local Government, Development/humanitarian interest, Environmental interest, Community or organized citizens, Cultural Interest

The Chinese Government originally estimated only 725,000 people to resettle, a deliberate underestimation to help the project's approval. In 1992, the authorities announced the total figure of resettlement was from 846,200 up to 1.2 million people in 2009. Independent sources raised this figure to 2 million and even more. Ultimately, more than 100 towns were submerged. Fourteen thousand hectares of agricultural land were expected to be submerged, as were more than 100 archeological sites, some dating back over 12,000 years. In late 2007, government announced that "at least 4 million people from the Three Gorges Reservoir area are to be relocated to cities in the next 10 to 15 years" It now appears possible that as many as 6 million people in total will have to be resettled because of the dam and surrounding impacts.[13]

About 43% of the people relocated are peasants who were to be compensated with new land to be farmed, but most of the land in the reservoir region is not able to be farmed because of its steeper slopes with very thin soils, which have been proved to be less fertile and unproductive, so there is not enough land to distribute.[14]

Chinese Government - President of China - Premier State Council - Vice President - Communist Party of China - National People's Congress (NPC) - Ministry of Water Resources - Yangtze Valley Planning Office - The Three Gorges Corporation - Provinces and municipalities local governments.

NGO's and Civil Society - International Rivers

- Human Rights Watch Asia

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Key Questions

Power and Politics: *To what extent can international actors and movements from civil society influence water management? How and when is this beneficial/detrimental and how can these effects be supported/mitigated?*

This project effected a range of actors who were not consulted or otherwise included in the decision making process . However, their activism related to the project outcomes has contributed to promote inclusion, to some extent. Also, technical expertise from the local scientific community and international best practices has contributed to tackle the immediate consequences of the TGD. The following paragraphs describe their activities and their struggle for a comprehensive solution to the problems posed by the TGD.

Public Demonstrations and Protest – In general public opinion was excluded and never consulted about the project, particularly those affected for flooding and resettlement. In May 1992, the police arrested 179 members of the Democratic Youth Party and charged them with counterrevolutionary activities for sabotaging the progress of the TGD. To this day, their whereabouts are still unknown. Human Rights Watch Asia has condemned repression from the State and appealed to foreign governments and corporations to assist in discovering the fate of the dissenters.

Scientific and Expert opinion – Critics in China from scientific, engineering and environmental backgrounds who have attempted to raise technical questions about the project have been accused of counterrevolutionary intent and disloyalty. Opponents have been persecuted since 1956. Criticism of the dam is strictly forbidden. Books which contain such debates are banned. Experts and journalists who attempted such discussions were harassed, even jailed. Li Rui, Mao's personal secretary and vice

minister of water resources was the first to be purged during the Great Leap Forward. Years later, in 1989 a journalist Dai Quing, published the book Yangtze! Yangtze! a collection of articles by several scholars against the dam. After the Tiananmen debacle her book was officially banned and 30,000 copies were destroyed. The authors were accused of provoking chaos and riots. Dai was imprisoned for 10 months. The eminent scientist Qian Jiaju, who openly criticized the project, was forced into exile. The controversy over the dam, which began out of genuine concern over the environmental dangers and future economic health of China, turned into political fight.[8]

Congress Representatives – Some have been prevented from expressing their opinions at meetings with the main government body planning the project. They have accused pro-dam Communist Party officials for employing unscientific methods and indulging impossible plans in their ambitions to build the world's biggest dam. Censorship and lack of open discussion increased uncertainty concerns and diminished trust on government decisions preventing mutual gains for the entire society. Anti-dam lobbyists believed that public access to basic information about the dam would have helped to prevent flood disasters.[15]

Research Institutions – Among other valuable attempts to implement follow-up efforts to obtain reliable information are those pursued by ecologist Chen Guojie and his colleagues at the Mountainous Disaster and Environment Institute in Chengdu and the Chinese Academy of Sciences (CAS) setting up seven experiment stations, one of them located at Wanxian, an area where some 800,000 residents are being relocated. The team monitored environmental changes during construction of the reservoir as well as the fate of the residents. Those stations feed into a broader network of several hundred scientists in 70 institutions who are part of a $110 million project to monitor a dozen aspects of the project's impact. Such initiatives remained isolated from mainstream politics and management of the TGD and just gained notoriety in the last decade with government shifting leadership and enhanced professionalization of Chinese public service.

Government officials and bureaucracy – Lack of accountability from local, provincial, and central government prompts collaboration failure. Corruption and absence of transparency mechanisms hinder adaptive management and effective rapport from impacted populations and government authorities. In early 2000 the Chinese government released information that corrupt officials had embezzled $60 million (500 million yuan) from resettlement funds for the TGD. An official was sentenced to death for embezzling almost $1.5 million from the project.[6]

The consequences of the TGD will continue to have significant impact in China and other countries (pollution and acid rain spreading towards South Korea and Japan). More actors (foreign and local) are involved to find solutions to their problems and make their claims be addressed. In terms of the decision making process some specialists argue that this has been a symbolic depiction of the power struggle between reformist and hardliners inside Chinese politics.[16] It has been an ongoing process, even after the dam completion and it will continue until adequate solutions are accomplished.

Power and Politics: ***To what extent can international actors and movements from civil society influence water management? How and when is this beneficial/detrimental and how can these effects be supported/mitigated?***

Chinese Authorities are hard bargainers and typically withhold information, keep secrecy around negotiations and favor a close decision-making process, but there are areas of opportunity in the enormous environmental problems China is facing nowadays, as a result of its rapid economic growth, mainly air pollution, land degradation, and water contamination. These pressing issues pose tremendous challenges for China's continued economic development, political stability, public health and sustainability.

In its early stages, the TGD sought for participation and involvement of Multilateral Financial Institutions such as the World Bank and the Asian Development Bank. Export Credit Agencies such as the US Export-Import Bank and other investment banks and corporations were invited. However, the Chinese government refused to comply with the World Bank standards in environmental and social policy issues (particularly resettlement), so the World Bank refused to finance the project. This made potential investors to hesitate about the viability of the project, and increased their uncertainty with the way the Chinese government would be dealing with problems related to technical and management issues, both for water management and power generation. Other agencies like the US Ex-Im Bank walked out the negotiations. But Chinese ambitions were not to be discouraged since growing export reserves and a booming economy allowed the government to finance the project through China's Development Banks.[17]

In past decades, the World Bank used to be a big advocate of hydropower generation, and promoted investments for their construction as a solution to air contamination and reduced CO2 emissions, but perceptions of dams and dam building have changed. Dams today symbolize, for some critics, not progress but environmental and social devastation.[18]

International pressures on environmental issues have found its way through international mechanisms aimed to provide standard regulations to the international community on critical issues as Agenda 21 emanated from the 1992 United Nations Conference on Environment and Development. In recent years, China has started to shift its economic development toward a more environmentally sustainable environment, stimulated by mechanisms such as Agenda 21 and other international agreements.</ref> In the year 2000, the World Bank sponsored a World Commission on Dams study entitled Dams and Development: A New Framework for Decision-Making. It was the first comprehensive global and independent review of the performance and associated impact of large dams. The document summarized most of the mea culpa, pinpointing the flaws, omissions and errors undertaken by large dam projects.

As the TGD project continued, international media began to report on growing threats from landslides, pollution, flooding, as well as growing social and political unrest and dissatisfaction with relocating millions of people. Foreign public opinion and China's international prestige play a significant role in the international scene, since Chinese foreign policy aims to lead the causes of the developing world. On that note, International Regimes have induced China to join multilateral efforts

such as the United Nations Conference on Environment and Development (UNCED), the "Earth Summit". As a result China incorporated some sustainable development policies in its Ninth Five-Year Plan (1996-2000), and the Fifteenth Chinese Communist Party Congress (1997).[19]

Multilateral negotiation forums typically offer mutual gain solutions with financial resources, technology transfer and important capacity-building (ex. Global Environment Facility) to entice participants and provide incentives to developing countries willing to address these problems. International negotiations have contributed to some remarkable adjustments on China's policies and actions in this issue area towards sustainability. International environmental agreements can effectively exercise some influence in public policy, providing guidance for reform and strengthening the underlying capacity of the agencies and actors involved.

Furthermore, Chinese investments have been actively supporting numerous infrastructure projects in Africa,[18] and increasing demands and critiques on Chinese practices in environmental issues have forced Chinese authorities to establish new domestic institutions and processes for managing those issues. These new institutions and processes favored environmental standards within China. For example, in 1995 the Air pollution Prevention and Control Law (first promulgated in 1987), was amended to include tougher regulations for controlling sulfur dioxide.[20]

International Non-Governmental Organizations such as International Rivers, WIN, M-Power, WWF, etc. also contribute to knowledge sharing and actively push for greater transparency and accountability on behalf of affected minorities and vulnerable groups, to advance their agenda on broader environmental concerns and other uncertainty related issues.

The energy infrastructure sector is increasingly being opened up to private investors, few of whom have proved willing to invest in a project that carries high risks and low returns. Financial institutions have embraced international standards such as the Equator principles, a benchmark developed by the financial industry to judge, evaluate and manage risk in project financing, including social and environmental impact for investment projects to perfect financial risk evaluation and early warning mechanisms.

These efforts combined with international regimes and international organizations such as United Nations Environmental Program – UNEP that promotes environmental standards and best practices. The International Hydropower Association's (IHA) launched in 2011 the industry's Hydropower Sustainability Assessment Protocol (HSAP) that enables dam builders to assess various sustainability topics, including biodiversity, erosion, infrastructure safety and downstream flow regimes. The protocol also includes guidelines on engaging with indigenous peoples and identifies the potential for reservoirs to produce greenhouse gas emissions.

The magnitude of the consequences of this controversial project has raised international concerns over appropriate water management and sustainability. All the initiatives discussed above emphasize mutual gains negotiations, joint fact finding, technical expertise, inclusion, transparency and collaborative adaptive management that are gradually improving governance in dam construction and water management in China.

eferences

1. ^ Marta Ponseti, Jordi López-Pujol and J. Waldo Villalobos, “The Three Gorges Dam Project in China: history and consequences” (2006), p. 157. Available at http://www.raco.cat/index.php/HMIC/article/viewFile/57768/67739 This paper analyses in retrospective the history, evolution and consequences of the TGD, the largest dam in the world, that this project entail in its design, construction, management and operation phases. It discuss the costs and benefits of the impact of the project in terms of flood control, power generation, navigation, water transfer, environment, biodiversity, archaeology, and resettlement.
2. ^ 2.0 2.1 2.2 Marta Ponseti, Jordi López-Pujol and J. Waldo Villalobos, “The Three Gorges Dam Project in China: history and consequences” (2006), p. 157. Available at http://www.raco.cat/index.php/HMIC/article/viewFile/57768/67739 This paper analyses in retrospective the history, evolution and consequences of the TGD, the largest dam in the world, that this project entail in its design, construction, management and operation phases. It discuss the costs and benefits of the impact of the project in terms of flood control, power generation, navigation, water transfer, environment, biodiversity, archaeology, and resettlement.
3. ^ 3.0 3.1 3.2 Peter H. Gleick, “Three Gorges Dam Project, Yangtze River, China,” World Water.org - The World’s Water 2008-2009-Waterbrief 3, p. 145. Available online from http://www.worldwater.org/data20082009/WB03.pdf
4. ^ Marta Ponseti, Jordi López-Pujol and J. Waldo Villalobos, “The Three Gorges Dam Project in China: history and consequences” (2006), p. 178. Available at http://www.raco.cat/index.php/HMIC/article/viewFile/57768/67739
5. ^ The original poem and its English translation can can be found in Zedong Mao, Willis Barnstone, and Ching-po Ko.(1972) Poems of Mao Tse-tung. New York: Harper and Row, pp 84- 87 (ISBN13: 9780835102575)) English Translation: “Great plans are afoot: A bridge will fly to span the north and south, turning a deep chasm into a thoroughfare; Walls of stone will stand upstream to the west to hold back Wuhan’s clouds and rain till a smooth lake rises in the narrow gorges. The mountain goddess if she is still there sill marvel at a world so changed”
6. ^ 6.0 6.1 Patricia Adams and Gráinne Ryder, “China's Great Leap Backward: Uneconomic and Outdated, the Three Gorges Dam Will Stunt China's Economic Growth,” International Journal , Vol. 53, No. 4 (Autumn, 1998), Canadian International Council, pp. 701-702. Available online from JSTOR: http://www.jstor.org/stable/40203722 A critique of an expensive, inefficient, gargantuan project which was authorized besides other low cost energy generation alternatives (cogeneration) have proven to be more efficient (even in China) to reduce Chinese reliance on coal as a source of energy. It analyzes Chinese political context and decision making framework and concludes on the high cost the dam will inflict on Chinese people, the environment and its economy.
7. ^ Marta Ponseti, Jordi López-Pujol and J. Waldo Villalobos, “The Three Gorges Dam Project in China: history and consequences” (2006), p. 174. Available at http://www.raco.cat/index.php/HMIC/article/viewFile/57768/67739
8. ^ 8.0 8.1 Audrey R. Topping, “Ecological Roulette: Damming the Yangtze,” Foreign Affairs , Vol. 74, No. 5 (Sep. - Oct., 1995), Council on Foreign Relations, pp.133 - 139 Available online from JSTOR: http://www.jstor.org/stable/20047305 This paper asks if the benefits of the TGD are worth the costs and the risks of the project. Her approach offers a contextual description of the events and the personalities of Chinese politics involved in the project. It also provide a social dimension of the effects on the livelihood of the people affected by the project, with examples or protest and opposition from the population including eminent researchers and journalists highlighting the immense human cost this project has created and the widespread violation of human and civil rights.

9. ^ From a Summary of Dai Miao et al (2011) Translation of A study on the relationship between water levels and seismic activity in the Three Gorges reservoir. Originally published September 2010 in the Journal of the Changjiang Water Resources Commission Volume 41, No. 17at http://probeinternational.org/library/wp-content/uploads/2011/06/3-Gorges-Report-26-5.pdf. Summary available at: http://journal.probeinternational.org/2011/06/01/chinese-study-reveals-three-gorges-dam-triggered-3000-earthquakes-numerous-landslides/
10. ^ Xiong Lei, “Going against the flow in China,” Science; Apr 3, 1998; 280, 5360; ProQuest SciTech Collection, pg. 24
11. ^ Marta Ponseti, Jordi López-Pujol and J. Waldo Villalobos, “The Three Gorges Dam Project in China: history and consequences” (2006), p. 173. Available at http://www.raco.cat/index.php/HMIC/article/viewFile/57768/67739
12. ^ **Cite error: Invalid `<ref>` tag; no text was provided for refs named `Ponseti 2006-178`**
13. ^ Marta Ponseti, Jordi López-Pujol and J. Waldo Villalobos, “The Three Gorges Dam Project in China: history and consequences” (2006), p. 173. Available at http://www.raco.cat/index.php/HMIC/article/viewFile/57768/67739
14. ^ Marta Ponseti, Jordi López-Pujol and J. Waldo Villalobos, “The Three Gorges Dam Project in China: history and consequences” (2006), p. 163. Available at http://www.raco.cat/index.php/HMIC/article/viewFile/57768/67739
15. ^ The World Commission on Dams, Dams and Development: A New Framework for Decision-Making (2000), p.187. http://www.internationalrivers.org/files/attached-files/world_commission_on_dams_final_report.pdf
16. ^ Lodge, George C., and Cate Reavis, "U.S. Export-Import Bank and the Three Gorges Dam (A),” The Harvard Business School Case 900-017, May 2000 (Revised July 2000).
17. ^ Stephanie Joyce, “Is It Worth a Dam?” Environmental Health Perspectives , Vol. 105, No. 10 (Oct., 1997), pp. 1050, Published by: Brogan & Partners. Available online from JSTOR: http://www.jstor.org/stable/3433845
18. ^ 18.0 18.1 Paul G. Harris and Chihiro Udagawa, “Defusing the Bombshell? Agenda 21 and Economic Development in China” Review of International Political Economy , Vol. 11, No. 3 (Aug., 2004), pp. 618-640 Taylor & Francis, Ltd. Available online from JSTOR: http://www.jstor.org/stable/4177513
19. ^ Peter Bosshard, “China Dams the World” World Policy Journal , Vol. 26, No. 4 (Winter, 2009/2010), Sage Publications Inc., pp. 43-51. Available online form JSTOR http://www.jstor.org/stable/40468737 This paper elaborates on China’s infrastructure projects in developing countries, exporting its experience with hydropower and its intent to “dam the world” promoting the construction of 220 dams in 50 countries, given that many developing countries cannot comply with international standard regulations (particularly in environment and social policy) and China seems to be the only source of development finance, leading to disastrous implementation in the developing world (mainly Africa) with questionable results and consequences for impoverished, disenfranchised people who will end up paying the price at a huge social cost.
20. ^ Yu Xiaogang And Ding Pin, “The Equator Principles and the Environmental Responsibilities of the Financial Industry In China,” in Dorothy-Grace Guerrero and Firoze Manji (eds), China's New Role in Africa and The South: A search for a new perspective, Fahamu Books and Pambazuka Press Publication (2008). Available online from: http://asiandrivers.open.ac.uk/Fahamu%20book%202008.pdf

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